

A High-Precision Positioning Servo Controller Based on Phase/Frequency Detecting Technique of Two-Phase-Type PLL

Takashi Emura, *Member, IEEE*, Lei Wang, *Member, IEEE*, Masashi Yamanaka, *Member, IEEE*, and Hisashi Nakamura

Abstract—This paper presents a new method proposed for high-precision positioning servomechanisms. The servo controller uses a two-phase-type phase-locked loop (PLL) to detect position tracking error and speed fluctuation with high resolution. Because the two-phase-type PLL has a wide frequency range and high noise suppression performance, we applied the new controller to high-power and high-velocity servo-spindles and achieved high control performance. The developed servo-spindles were used by a high-productivity numerically controlled gear grinding machine. Experiments of gear grinding were carried out, and the results confirmed the performance of the developed controller.

Index Terms—Machine tool control, motion control, phase-locked loops, position control, servo systems.

I. INTRODUCTION

HIGH-VELOCITY and high-precision servomechanisms are always an important affair for some numerically controlled (NC) machines. For example, an NC gear grinding machine needs servo-spindles of rotary velocity up to 2000 r/min for the work spindle, and up to 12 000 r/min for the grinding spindle [1]. Position error is only allowed tens of sec of arc¹ for the work spindle and hundreds of sec for the grinding spindle. Moreover, the grinding spindle has to rotate at a velocity of below 30 r/min for tool truing. Problems, such as how to detect the rotary angle of high-velocity rotation precisely and how to make a high loop gain, are difficult to solve.

The use of a phase-locked loop (PLL) in motor control has made progress for high-precision servomechanisms [2]–[4]. However, the conventional PLL systems cannot be used at very low velocity, because most of them use digital phase detectors [4]. An analog-type PLL system was also reported [5], but the response speed of the phase detector does not seem very high. All of these systems have no phase reversibility and their operation range is narrow. They are not suitable for high-precision positioning control.

A two-phase-type PLL method [6], which has a wide operation range and phase reversibility, was tried to solve these problems [7]. Since no low-pass filter is necessary for the phase detector of the two-phase-type PLL, the response characteristic is significantly improved.

The two-phase-type PLL method was first proposed by Emura and has been utilized in several applications [1], [7]–[9]. A high-precision extremely low-velocity rotary table was achieved by the two-phase-type PLL [7], and high-precision servo-spindles were developed for NC gear grinding machines basically depending on this method [1], [8]. Moreover, this method was also applied to a high-resolution interpolator for incremental encoders of wide dynamic range [9].

A high-velocity and high-precision servo-spindle reported in [8] used a two-phase-type PLL for positioning control. Its servo controller had an inner velocity feedback loop to compensate loop stability. Two rotary sensors were used. One was for positioning control and the other for inner velocity feedback loop. A steady-state error was caused by the velocity feedback loop, and for this reason, feedforward compensation was tried to cancel this steady-state error. However, a two-sensor system is not always available and the velocity feedback compensation has to use the same sensor with positioning control in most cases. Another disadvantage of this method is that the control accuracy probably becomes unstable due to parameter drift of the feedforward compensator and inner velocity feedback loop.

The positioning servo controller of the two-phase-type PLL proposed in this paper uses a two-phase-type PLL for detection of position error and speed fluctuation. It utilizes the excellent capability of the PLL for phase tracking and frequency tracking with high noise-rejection ability [10], [11]. With the servo error detector, the position error and speed fluctuation can be detected precisely. The authors used the servo error detector of the two-phase-type PLL in a high-precision positioning servo controller. This method enables a servo system to achieve very large damping without steady-state error, whereby its control accuracy becomes very stable.

A high-precision positioning servo controller of the two-phase-type PLL was developed by the authors and was used to control a servo-spindle. The servo motor of the servo-spindle is of a rated power of 20 kW and a rotary velocity range from tens of revolutions per minute up to 10 000 r/min. This servo-spindle was applied to a productive-type NC gear grinding machine that was developed for automobile production lines. In this paper, the servo error detector using two-phase-type PLL is presented, and a high-velocity and

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T. Emura and L. Wang are with the Department of Mechatronics and Precision Engineering, Tohoku University, Sendai 980-8579, Japan (e-mail: wang@emura.mech.tohoku.ac.jp).

M. Yamanaka is with the Department of Machine Intelligence and Systems Engineering, Tohoku University, Sendai 980-8579, Japan.

H. Nakamura is with Toyoda Machine Works, Ltd., Kariya 448-8652, Japan. Publisher Item Identifier S 0278-0046(00)10266-7.

¹“sec” is used to stand for seconds of rotary angle and “s” for seconds of time in this paper.

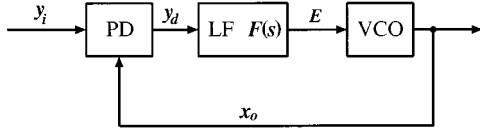


Fig. 1. Block diagram of ordinary PLL.

high-precision servomechanism is discussed. In addition, experimental results of the servo-spindle and the NC gear grinding machine are given to show the performance of the proposed method in practical applications.

II. TWO-PHASE TYPE PLL METHOD

PLL and its applications in motor control have been studied widely. The proposed servo controller uses it to detect position error and speed fluctuation. In order to make the discussion easier, a brief review of the PLL and two-phase-type PLL is introduced.

A. Principle of PLL

Generally, a PLL contains three basic components: a phase detector (PD), a loop filter (LF), and a voltage-controlled oscillator (VCO) as shown in Fig. 1 [10]. Notations for Fig. 1 are as follows:

- y_i input signal;
- x_o output signal of the VCO;
- y_d output of the phase detector;
- $F(s)$ transfer function of the loop filter;
- E output of the loop filter.

Supposing a sinusoidal phase detector is used, the input signal and the output signal of the VCO are required as follows:

$$y_i = a \sin \theta_i \quad (1)$$

$$x_o = b \cos \theta_o \quad (2)$$

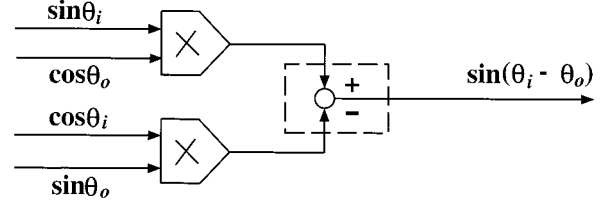
where θ_i and a are the phase angle and amplitude of input signal, respectively, and θ_o and b are those of the output signal of the VCO. The phase detector operates as a multiplier and performs the following function:

$$y_d = a \sin \theta_i b \cos \theta_o = \frac{1}{2} ab \{ \sin(\theta_i - \theta_o) + \sin(\theta_i + \theta_o) \}. \quad (3)$$

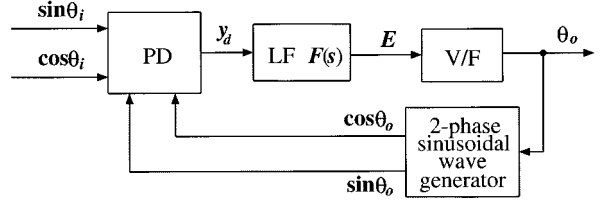
If the loop is locked, the VCO oscillates at the same frequency as the input signal. The first term of (3) has a low frequency and the second one has a high frequency. The PLL needs a low-pass filter to remove the high-frequency term $\sin(\theta_i + \theta_o)$. If the phase difference $\theta_i - \theta_o$ is kept small, (3) can be approximately represented as

$$y_d = \frac{1}{2} ab(\theta_i - \theta_o). \quad (4)$$

When the phase difference exceeds $\pm 90^\circ$, the loop would cause cycle slip or fall out of lock. In communication instruments, a PLL is used for phase or frequency discrimination with high noise immunity [11]. It is also used in motor control by locking a motor to a reference input signal [2].



(a)



(b)

Fig. 2. Two-phase-type PLL. (a) Phase detector for two-phase signal. (b) Schematic of two-phase-type PLL.

B. Two-Phase-Type PLL Method

A low-pass filter is necessary for the ordinary PLL to remove the high-frequency term. For an input of limited velocity range, exact phase difference can be extracted by setting the cutoff frequency of the loop filter sufficiently low in comparison with input frequency. However, for an input of wide velocity range, this method is less effective, because when the cutoff frequency has to be decided on a very low frequency to meet a low-frequency input, it is not able to respond to a high-frequency input. On the other hand, a motor is probably required to have a very wide velocity range, as mentioned in Section I. In particular, it has to be controlled precisely, even when the motor stops and changes rotary direction. In the case of reversal of rotation, the sign of ω_i ($=\dot{\theta}_i$) changes. This means that ω_i becomes zero instantaneously. Thus, the ordinary PLL cannot be used in cases where phase reversibility is required, because the cutoff frequency of the loop filter cannot be set to zero.

1) *Phase Detector for Two-Phase-Type PLL*: This problem was solved by the two-phase-type PLL because it does not need a low-pass filter for its phase detector. Fig. 2(a) shows the phase detector used for the two-phase-type PLL that contains two multipliers and one subtractor. This phase detector requires two-phase signals of instruction

$$\begin{cases} x_i = \cos \theta_i \\ y_i = \sin \theta_i \end{cases} \quad (5)$$

and two-phase signals of output from the VCO

$$\begin{cases} x_o = \cos \theta_o \\ y_o = \sin \theta_o \end{cases} \quad (6)$$

for its input, where signals are supposed to have been normalized. The operation of this phase detector can be explained with rotating phasors, when the two-phase sinusoidal signals are considered as real and imaginary components of a rotating phasor. The imaginary component of the phasor of phase difference is calculated by

$$y_d = \text{Im}[e^{i\theta_i} e^{-i\theta_o}] = \sin(\theta_i - \theta_o). \quad (7)$$

Comparing (7) with (3), it is clear that there exists no high-frequency component at the output of the two-phase-type phase detector. Therefore, a two-phase-type phase detector does not need a low-pass filter to remove the high-frequency component.

2) *Principle of Two-Phase-Type PLL Method:* Fig. 2(b) shows a schematic drawing of the basic structure of the two-phase-type PLL [6]. This loop uses a two-phase-type phase detector. The VCO for this loop has a structure which generates two-phase sinusoidal waves at a frequency proportional to input voltage. The phase difference calculated by the phase detector is applied to the loop filter. The output of the loop filter is used for controlling the VCO so as to make the phasor of the VCO track the instruction phasor.

A loop filter is needed to compensate loop dynamics as well. However, because the loop filter is not required by the phase detector to remove a high-frequency component, parameters of the loop filter can be determined to meet the requirement specifications imposed on loop dynamics, noise rejection, irrespectively of input frequency. This feature makes obtaining a wide dynamic range easy. The two-phase-type PLL offers an apparent advantage when it is intended to control high-precision positioning servomechanisms.

III. SERVO CONTROL USING TWO-PHASE-TYPE PLL METHOD

Use of a PLL in motor control for high accuracy has been widely researched. However, applications of a PLL were almost always limited to systems of constant-velocity control. This is because the conventional PLL has a very limited width of operation range and phase reversibility is not available, whereas the two-phase-type PLL has a wide operation range and phase reversibility, which are necessary for high-precision positioning servo controllers.

A. Motor Control Using Two-Phase-Type PLL

A motor control system of a wide velocity range can simply be constructed by substituting the VCO with a motor and an encoder. The two-phase sinusoidal waves can be generated by the encoder of an optical or magnetic type. This is not difficult because the original output of such encoders usually contains two-phase sinusoidal or nearly sinusoidal waves before being converted into square-wave pulses by waveform-shaping circuits. As no low-pass filter and quantization are required by the phase detector, the motor can be controlled from a very low velocity to a high one and even be controlled in the case of reversal of motor rotation.

Although velocity control of wide range is available, it is difficult to use for high-precision positioning servomechanisms. This is because conventional controllers of a PLL usually use phase detectors to detect phase and frequency difference and treat them as servo error: position error or speed error. The lock-in range of such systems is less than one wavelength of input sinusoidal waves. On the other hand, high-precision positioning servomechanisms usually employ high-resolution encoders. The wavelength of the encoder output signals becomes very small compared with the shaft mechanical angle. This makes the PLL skip cycle or lose lock very easily, which means the failure of the positioning control.

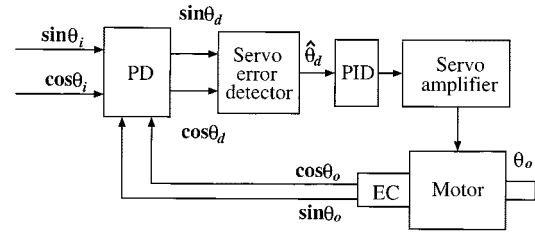


Fig. 3. A positioning servo system of two-phase-type PLL.

B. Positioning Servo Control Using Two-Phase-Type PLL

The controller of the PLL for high-precision positioning servo control is required to be robust so that no cycle slip or loss of lock takes place in any case. However, under such conditions as varying instruction input, load disturbances, or strong noise, it is difficult for a conventional PLL to maintain a locked state. A large lag angle occurs easily under a varying instruction input. This is because a motor has much low response speed due to its large mechanical inertia compared with a VCO implemented by electronic circuits, while, under a large load disturbance a large phase difference will take place. Moreover, strong noise perhaps makes misoperation of the phase detector a very high probability. For these reasons, a servo error detecting method with a wide hold-in range and tracking performance is required.

These requirements can be met by using the two-phase-type PLL for the detection of servo error. The structure of a positioning servo system of a two-phase-type PLL is schematically shown in Fig. 3, wherein a servo error detector is embedded. This feedback loop is to control the phasor associated with the encoder to track the instruction phasor. The tracking error is represented by the phase difference of the two phasors $\theta_d = \theta_i - \theta_o$, and is used as the servo error for positioning control. The phase difference θ_d is detected by using the servo error detector of the two-phase-type PLL. For this purpose, the phase difference θ_d is required to be converted to two-phase sinusoidal waves as follows:

$$\left. \begin{aligned} x_d &= \cos \theta_d \\ y_d &= \sin \theta_d \end{aligned} \right\} \quad (8)$$

which is referred to as the position error phasor in this paper. In order to detect the servo error, we only need to detect the phase of the position error phasor. For this purpose, the servo error detector generates a pseudophasor in the VCO. This pseudophasor is controlled to lock on the position error phasor by the two-phase-type PLL. The locked state is easily kept because the VCO in the servo error detector is implemented with electronic circuits. For the same reason, the phase and frequency associated with the pseudophasor, which are the system estimates of servo the error, are easily obtained from the VCO and are used to control the servo motor.

This servo error detector transfers the servo error to a voltage signal proportionally and irrespectively to how large the phase error is. Therefore, the positioning servo system of the two-phase-type PLL is not out of lock, even with the existence of disturbances or change in velocity instruction. The PLL, as is known, has excellent ability in phase and frequency discrimi-

nation, whereby servo error, including position error and speed fluctuation, is obtained precisely.

1) *Servo Error Detection Using Two-Phase-Type PLL*: The structure of a servo error detector using a two-phase-type PLL is shown in Fig. 4(a), where the pseudophasor generated by the VCO can be represented as follows:

$$\begin{cases} \hat{x}_d = \cos \hat{\theta}_d \\ \hat{y}_d = \sin \hat{\theta}_d \end{cases} \quad (9)$$

where $\hat{\theta}_d$ is the system estimate of position error. Similarly to (7), the following result is available:

$$E_d = \sin(\theta_d - \hat{\theta}_d) \approx \theta_d - \hat{\theta}_d \quad (10)$$

where E_d is the tracking error of the servo error detector. The VCO is a composite of a V/F converter and an up/down binary counter. The V/F converter generates a pulse train with its frequency in proportion to input voltage. The pulses are then fed to the counter. The results of the counter are converted to two-phase sinusoidal waves by a sinusoidal wave generator. This can be done by using results of the counter to decode ROMs in which sin and cos data are stored. Subsequently, $\hat{\theta}_d$ is available in digital form from the counter or in analog form by using an A/D converter.

The servo error detector transfers sinusoidal signal to phase signal linearly with an arbitrarily large output range. This is achieved by using a counter of sufficient bit length to count pulses from the V/F converter. The up/down binary counter has n bits, for instance, and its low k bits are used for the waveform generator. The counter can accumulate a maximum of 2^n counts of pulses without overflow, while one period of the sinusoidal waves is generated by 2^k counts of pulses. This means that the servo error detector can deal with 2^{n-k} periods of sinusoidal waves. Thus, a wide detection range of 2^{n-k} wavelength of sinusoidal waves can be obtained. The bit length of a counter can be set arbitrarily long, hence, a very wide detection range can be achieved. By this method, high-resolution detection is readily obtained. The interpolation rate is 2^k , that is, it divides one sinusoidal wave into 2^k pulses.

2) *Speed Fluctuation Detection Using Two-Phase-Type PLL*: A precise speed fluctuation signal can be obtained by using the servo error detector as well. From Fig. 4(a), it is observed that, when the loop is locked, that is, $\theta_d - \hat{\theta}_d \rightarrow 0$, $\hat{\omega}_d (= \dot{\hat{\theta}}_d)$ will approach $\omega_d (= \dot{\theta}_d)$. For a constant ω_d , there exists no tracking error in $\hat{\omega}_d$, that is, no steady-state error for detection of speed fluctuation using the servo error detector. The signal of $\hat{\omega}_d$ is available either in analog voltage from the output of the loop filter or pulse train from the V/F converter depending on the convenience of the controller. In the case of speed fluctuation detection, the noise rejection performance and high resolution of the PLL are of importance to high-precision servomechanisms.

3) *Characteristics of Servo Error Detector*: Fig. 4(b) is a block diagram of the servo error detector, where K_{pd} is the gain of the phase detector, and $K_{V/F}$ is the transfer coefficient of the V/F converter. k is the bit length of the binary counter used for

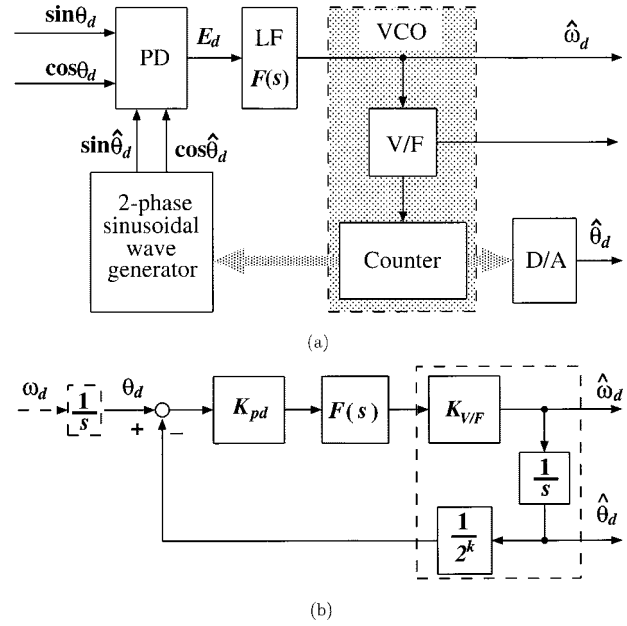


Fig. 4. Servo error detection using two-phase-type PLL. (a) Servo error detector using two-phase-type PLL. (b) Block diagram of servo error detector.

the sinusoidal wave generator. If the PLL for the servo error detector is locked, its transfer function can be obtained as follows:

$$G_e(s) = \frac{\hat{\theta}_d}{\theta_d} = \frac{2^k K_{pd} K_{V/F} F(s)}{2^k s + K_{pd} K_{V/F} F(s)}. \quad (11)$$

For example, if a lead-lag filter is used, as is usually the case, its transfer function is given by

$$F(s) = K_d \frac{1 + \tau_2 s}{1 + \tau_1 s} \quad (12)$$

where τ_1 and τ_2 are the time constant, and K_d is gain associated with the loop filter. Therefore, we have

$$G_e(s) = \frac{2^k \omega_n^2 (1 + \tau_2 s)}{s^2 + 2\xi \omega_n s + \omega_n^2} \quad (13)$$

where

$$\omega_n = \sqrt{\frac{K}{2^k \tau_1}} \quad \xi = \frac{2^k + K \tau_2}{2\sqrt{2^k K \tau_1}} \quad (14)$$

are the natural frequency and the damping ratio of the servo error detector, respectively, and $K = K_{pd} K_{V/F} K_d$ is the total loop gain of the servo error detector. As the loop filter of the two-phase-type PLL is decided irrespective of input frequency, the dynamics can be designed due to required noise rejection, hold-in, and tracking performance. Nevertheless, tracking performance requires wide frequency-response bandwidth, and the noise rejection of the PLL is distinguished compared to the conventional method of servo error detection for high-power servo systems, where power amplifiers usually use a switching power device to drive servomotors. Large amplitude switching noise is present at the encoder output.

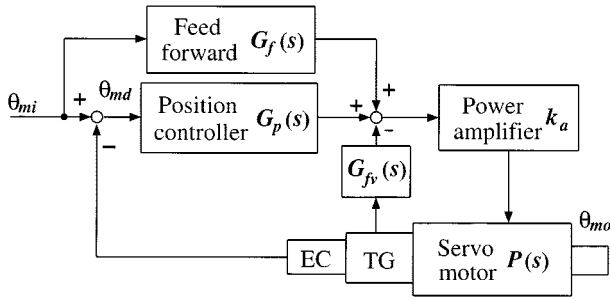


Fig. 5. Feedforward compensation.

C. Servo Control Using Two-Phase-Type PLL

High-precision servo systems always employ high loop gain method. High tracking precision and robustness to load disturbance are able to be obtained by this method [12]. However, the stability of a servo system is probably deteriorated by an increase of loop gain because of the inadequacy for damping. In particular, a large damping ratio of the motor is not expected for the decrease of power lost. Thus, positioning servomechanisms use inner velocity feedback loops for stabilizing the systems. An inner velocity feedback loop gives rise to a steady-state error, called a residual velocity error, in a servo system. This error can be canceled by using feedforward compensation.

Fig. 5 illustrates schematically a typical structure of servo systems using feedforward compensation, where θ_{mi} is the instruction angle, θ_{mo} is the rotation angle of the motor and θ_{md} is the angular position error, that is, $\theta_{mo} = \theta_{mi} - \theta_{mo}$. Let the transfer function of the inner velocity feedback loop

$$G_v(s) = \frac{k_a P(s)}{1 + G_{fv}(s) k_a P(s)} \quad (15)$$

where

- $P(s)$ transfer function of a motor shaft;
- k_a torque constant of the motor;
- $G_{fv}(s)$ transfer function of the velocity feedback compensator.

Moreover, the motor driver is considered as an ideal current-controlled power amplifier whose transfer function equals unity. The transfer function $S_{ff}(s)$ from θ_{mi} to θ_{md} is obtained as follows:

$$S_{ff}(s) = \frac{1 - G_f(s)G_v(s)}{1 + G_p(s)G_v(s)} \quad (16)$$

where $G_f(s)$ is the transfer function of the feedforward compensator, and $G_p(s)$ is the transfer function of the positioning control compensator, thus the term of $G_f(s)G_v(s)$ in the numerator is dependent on the feedforward compensator. Substituting (15) into (16) yields the following:

$$S_{ff}(s) = \frac{(1 + G_{fv}(s)k_a P(s)) - G_f(s)k_a P(s)}{1 + (G_p(s) + G_{fv}(s))k_a P(s)}. \quad (17)$$

² θ indexed by subscript m indicates a mechanical shaft angle, while θ is for an electrical phase angle. They have a simple relation $\theta = n\theta_m$, where n is the number of slits of an encoder.

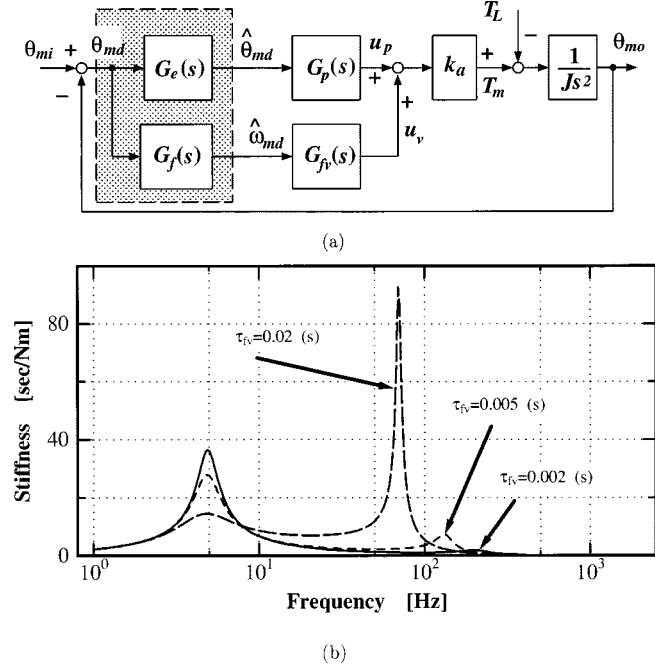


Fig. 6. High-precision servo control. (a) Block diagram of positioning servo system of two-phase-type PLL using phase/frequency detecting method. (b) Load disturbance characteristic versus τ_{fv} , where $K_p = 2189.199$ V/rad, $K_{fv} = 294.94$ V/(rad/s), $k_a = 21.2$ N·m/V, $J = 0.1852$ kg·m², and $K_I = 295$ s.

Simply, letting $G_p(s) = K_p$, $G_f(s) = K_f s$, $G_{fv}(s) = K_{fv} s$ and $P(s) = 1/(Js^2)$, the steady-state position error is

$$\lim_{s \rightarrow 0} \theta_{md} = \frac{K_{fv} - K_f}{K_p} \omega_{mi} \quad (18)$$

where $\omega_{mi} (= \dot{\theta}_{mi})$ is the velocity of instruction, K_p , K_f , and K_{fv} indicate the gain of the positioning control compensator, the feedforward compensator, and the velocity feedback compensator, respectively, and J is the motor inertia. If there is no feedforward compensator, that is, $K_f = 0$, there is residual velocity error $\omega_{mi} K_{fv} / K_p$. In the feedforward compensation, by tuning the feedforward compensator and the velocity feedback loop, K_f is made to equal K_{fv} , whereby it is able to reduce the steady-state error to a very small value. Even though motor friction may induce error to (18), it has been omitted because it is very small compared with the velocity feedback.

However, stable control precision is difficult to obtain. Parameter drift may destroy the balance of the matched feedforward compensator and velocity feedback loop. This introduces an unallowably large steady-state position error unavoidably into the servo systems.

This problem does not occur if the servo error detector discussed above is used. Its block diagram is shown in Fig. 6(a). This controller uses the speed fluctuation signal detected by the servo error detector to compensate system dynamics. As with a feedforward compensator, no residual velocity error is brought into the system by the compensation. The transfer function $S_{pd}(s)$ from θ_{mi} to θ_{md} is obtained as follows:

$$S_{pd}(s) = \frac{1}{1 + (G_p(s) + G_{fv}(s))k_a P(s)} \quad (19)$$

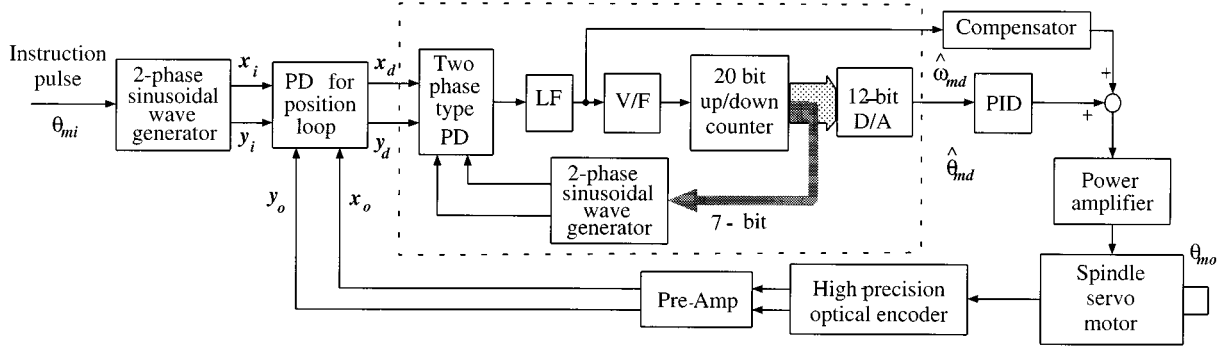


Fig. 7. Positioning servo system based on phase/frequency detecting technique of two-phase-type PLL.

where the approximation $G_e(s) = G_f(s) = 1$ is used for the reason that the natural frequency ω_n of the servo error detector is much higher than that of the servo system. Because

$$\lim_{s \rightarrow 0} \theta_{md}(s) = \lim_{s \rightarrow 0} S_{pd}(s) \theta_{mi}(s) = 0 \quad (20)$$

no residual velocity error is present. Because no parameter drift and adjustment are needed, the stability of the control precision is able to be improved significantly.

The noise rejection performance in speed fluctuation detection is of importance for control performance. For example, the transfer function from T_L to θ_{md} is obtained as

$$G_L(s) = \frac{P(s)}{1 + (G_p(s) + G_{fv}(s))k_a P(s)} \quad (21)$$

which represents the load disturbance suppression characteristic or stiffness. Let $P(s) = 1/(Js^2)$ and we have

$$G_p(s) = K_p \left(1 + \frac{K_I}{s} \right) \quad G_{fv} = \frac{K_{fv}s}{1 + \tau_{fv}s} \quad (22)$$

where K_I is the gain of the integrator, and τ_{fv} is the time constant of a low-pass filter used to remove deviation noise. Equation (21) is rewritten as

$$G_L(s) = \frac{1}{Js^2 + \frac{K_{fv}k_a}{(1 + \tau_{fv}s)}s + K_p k_a + \frac{K_I K_p k_a}{s}} \quad (23)$$

which is plotted in Fig. 6(b). τ_{fv} has to be very large in a conventional controller such as a proportional–integral–derivative (PID) controller because there may be very strong noise in the detected servo error signal. Therefore, a large feedback gain cannot be obtained because the position error may become large in the high-frequency range, as shown in Fig. 6(b).

From the above discussion, it can be concluded that positioning servo control based on the two-phase-type PLL can use another two-phase-type PLL to detect servo error for the purpose of high performance. The precise servo error is available based on the capability of the PLL for phase-frequency discrimination. An arbitrary hold-in range for the servo system is possible. A wide operation range can be obtained by wide dynamic range and phase reversibility of the two-phase-type PLL.

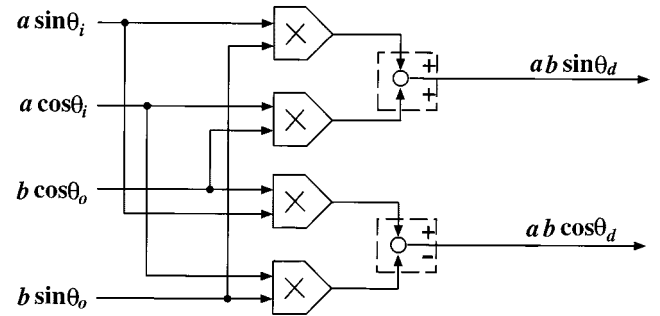


Fig. 8. Phase detector for position loop.

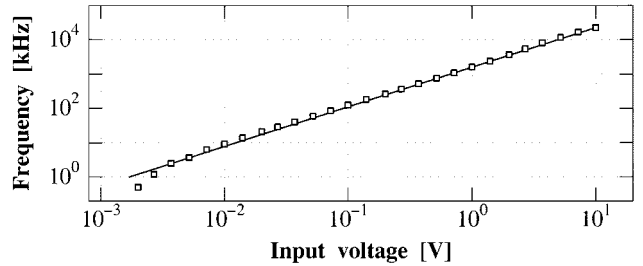


Fig. 9. Dynamic range of VCO.

IV. STRUCTURE OF POSITIONING SERVO CONTROLLER OF TWO-PHASE-TYPE PLL USING PHASE/FREQUENCY DETECTING TECHNIQUE

The servo controller of the two-phase-type PLL makes use of the phase/frequency detection capability of the PLL to realize high-precision positioning servomechanisms. The outline of a servomechanism using the developed positioning servo controller is illustrated schematically in Fig. 7. This controller uses the servo error detector of the two-phase-type PLL to detect position error and speed fluctuation, whereby it can control a servo motor.

A pulse train with up/down signal or up/down pulse trains is used for position or velocity instruction. The pulses themselves are the position instruction and their frequency is the required rotary velocity. The up/down signal controls the direction of motor rotation. The instruction pulse train is accumulated by a reversible binary counter and converted to two-phase sinusoidal waves by a two-phase sinusoidal wave generator. The two-phase sinusoidal wave generator contains two ROMs and

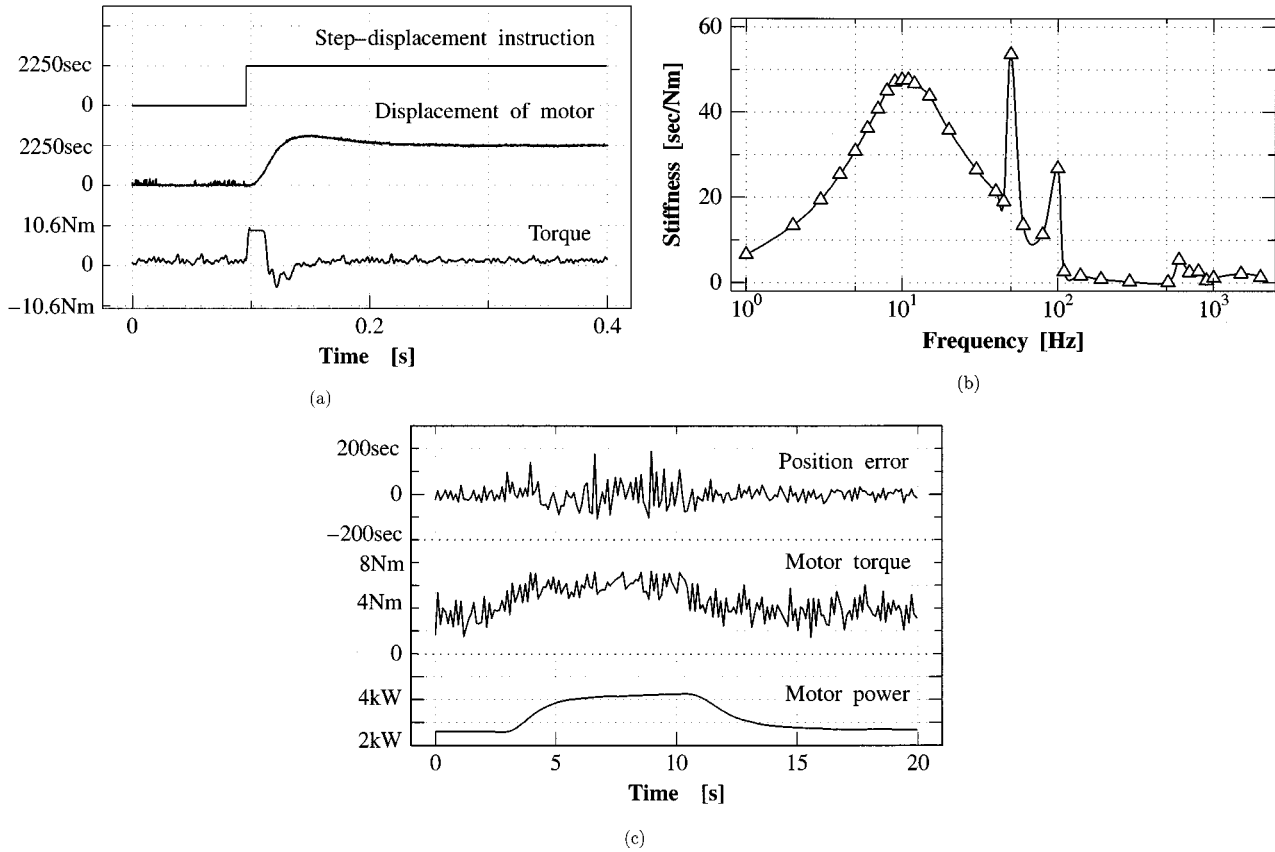


Fig. 10. Experimental results. (a) Step response of servo spindle. (b) Load characteristic of servo spindle. (c) Behavior of servo spindle in gear grinding process.

two D/A converters. The sin and cos data were stored in the ROMs with a bit length, e.g., m bit. When the ROMs are decoded by the output of the counter, digital sin and cos waves are available. They are then converted to analog sin and cos waves by the D/A converters. If the frequency of instruction pulse train is f_i Hz, two-phase sinusoidal waves of $f_i/2^m$ Hz are obtained.

A high-precision encoder is used, from which two-phase sinusoidal waves are obtained. Because the servomotor is controlled to make the sinusoidal waves of the encoder track instruction sinusoidal waves, the rotary velocity of the motor will be $f_i/2^m/R_m$ r/s, where R_m is encoder line density, i.e., R_m pairs of two-phase sinusoidal waves are available per shaft revolution. Thus, the minimum step position instruction of one pulse is $1\ 296\ 000/R_m/2^m$ sec.

The phase detector for position loop calculates the position error phasor from the instruction phasor and the output phasor of the shaft encoder. This can be done by using four multipliers, one adder, and one subtractor, as illustrated in Fig. 8. The phase detector for position loop carries out operations as follows:

$$\begin{aligned} x_d &= k_m(x_i x_o + y_i y_o) \\ &= k_m(a \cos \theta_i b \cos \theta_o + a \sin \theta_i b \sin \theta_o) \\ &= abk_m \cos \theta_d, \end{aligned} \quad (24)$$

$$\begin{aligned} y_d &= k_m(y_i x_o - x_i y_o) \\ &= k_m(a \cos \theta_i b \sin \theta_o - a \sin \theta_i b \cos \theta_o) \\ &= abk_m \sin \theta_d \end{aligned} \quad (25)$$

where k_m is the gain of multipliers.

The resolution of the positioning controller is dependent on the interpolation rate of the servo error detector. For example, the bit length used for decoding ROMs is $k = 7$, that is, the interpolation rate is 2^7 , and the resolution of positioning will be $1\ 296\ 000/R_m/2^7$ sec.

A wide hold-in range for the positioning servo controller can be obtained by using a servo error detector with a wide detection range. The up/down binary counter in Fig. 7 has $n = 20$ bits and, for instance, the detection range is 2^{20} bits. This means a wide hold-in range of $2^{20-7} = 8192$ wave length of sinusoidal waves can be obtained. The bit length can be set arbitrarily long, therefore, a very wide hold-in range for the servo system can be achieved, so that the servo system does not experience loss of lock.

A PID controller for position loop compensation uses the position error to control the servomotor. A compensator uses the speed fluctuation signal to compensate loop dynamics, which was discussed earlier.

V. EXPERIMENTS

Servo-spindles were developed based on the positioning servo controllers as shown in Fig. 7 for a high-productivity NC gear grinding machine. This machine is to be applied to automobile production lines. Two servo-spindles were used there and they have to be synchronized very precisely in rotary angles. One of the two servo motors has a rated power of 20 kW and maximum speed of 10 000 r/min. The rated torque is 22 N·m. Its inertia is 0.1852 kg·m². The spindle was required

to have a working velocity range from below 30 r/min to a maximum velocity of 10 000 r/min with high-precision positioning. Its resolution is required to be higher than 20 sec. IReversal of rotation is also required. An optical encoder is mounted to the spindle and 1800 pairs of sinusoidal waves can be obtained per revolution; its pitch error is less than ± 3 sec. The servomotor is driven by a pulsewidth modulation (PWM) current-controlled power amplifier. Both the switching frequency of the inverter and the update rate of the current controller are 25 kHz. The bandwidth of the current control is higher than 2.4 kHz. The use of a standard position controller has been tried, but it could not track the position instruction properly because of the strong noise from the PWM power amplifier.

A large lag time perhaps occurs due to the large mechanical inertia of the motor. For example, the maximum acceleration of the spindle is $22/0.1852 = 119 \text{ rad/s}^2$. It takes 0.441 s to change rotary velocity from 0 to 500 r/min, so that when such step-changed velocity input is applied, a lag phase of $3309 \times 2\pi$ rad will occur. Because the detection range of servo error is 8192 of wave length, $8192/1800 = 4.55$ turns of shaft, it can withstand a step change of 787 r/min in the reference speed. To allow a larger step change, it is only necessary to increase the number of bits of the counter shown in Fig. 7.

The resolution of the servomechanism is $1296000/1800/2^7 = 5.625$ sec. The V/F converter used in the servo error detector is required to have sufficiently wide bandwidth of oscillation frequency. A newly designed V/F converter has a linear dynamic range of 86 dB from the minimum frequency of 1 kHz to the maximum frequency of 22 MHz, as shown in Fig. 9. This is necessary to guarantee wide dynamic range and tracking ability of the servo system. In the implemented servo error detector, $\tau_1 = 0.016$ s, $\tau_2 = 1.6 \times 10^{-5}$ s, $K_d = 100.24$, $K_{V/F} = 2.2$ MHz, and $K_{pd} = 10 \text{ V/rad}$. According to (14), $\omega_n = 18.4 \text{ kHz}$ and $\xi = 0.928$. Thus, the servo error detector has sufficient high response frequency in comparison to the servomechanism.

Experiments were carried out, and results are shown in Fig. 10. Fig. 10(a) shows the response of the servo-spindle to a step displacement input. Although the step displacement instruction is over $2250/(128 \times 5.625) = 3$ cycles of sinusoidal waves, no unlock or cycle slip occurred and the servo error is only 40 sec after settled. An overshoot of 2.8 dB is observed, which meets the 3-dB requirement of the servo-spindle. If required, the servo controller can operate with no overshoot by tuning the parameters such as K_p and F_{fv} .

The dynamic stiffness as a measure of load characteristic of the servo-spindle is shown in Fig. 10(b), where $\tau_{fv} = 0.002$ s. The difference from Fig. 6(b) is observed, which might be due to the approximation of the transfer functions and error in parameters. The picks at 50 and 100 Hz were caused by misoperation of the power amplifier, which can be easily removed in practice. Compared with Fig. 6(b), the pick at 600 Hz is significantly suppressed, which proves the effectiveness of damping compensation of speed fluctuation feedback.

The developed servo-spindle was used for gear grinding. High power and high torque were needed in this process. The behavior of the servo-spindle in the gear grinding process is shown in Fig. 10(c). Motor power increased to above 4 kW and motor torque was up to 7 N·m during the grinding process. The

servo error was kept below ± 200 sec. Moreover, no misoperation such as cycle slip was observed during the grinding process.

VI. CONCLUSIONS

In this paper, a high-precision servomechanism was developed using a two-phase-type PLL. A servo error detector using a two-phase-type PLL was used in the controller. This servo error detector is capable of detecting position error and speed fluctuation with high precision and high resolution. This is based on the excellent capability of the PLL in phase/frequency discrimination, wide operation range, and phase reversibility of the two-phase-type PLL. High performance of the positioning servomechanism of the two-phase-type PLL was achieved by using phase/frequency detecting technique. This was proved by experimental results. Moreover, it was also confirmed by a practical application of a high-productivity NC gear grinding machine.

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Takashi Emura (M'94) received the M.S. and Ph.D. degrees from Tohoku University, Sendai, Japan, in 1966 and 1969, respectively.

From April 1969 to March 1971, he was a Lecturer at Yokohama National University, Yokohama, Japan. In April 1971, he joined the Department of Precision Engineering (currently the Department of Mechatronics and Precision Engineering), Tohoku University, as a Lecturer. He became an Associate Professor in 1974 and a Professor in 1985. His research interests are servomechanisms for high-precision NC machines, control of legged and wheeled robots, and car electronics.



Lei Wang (M'94) received the B.E. degree from DaLian Institute of Science and Technology, DaLian, China, and the M.E. and Ph.D. degrees from Tohoku University, Sendai, Japan, in 1985, 1989, and 1992, respectively.

In 1992, he joined Tohoku University, where he is currently an Assistant Professor in the Department of Mechatronics and Precision Engineering. His current research interests are mechatronics, NC machines, servomechanisms, robotics, car electronics, and intelligent vehicles.



Hisashi Nakamura received the M.E. degree from Hokkaido University, Sapporo, Japan, and the Ph.D. degree from Tohoku University, Sendai, Japan, in 1985 and 1995, respectively.

In 1985, he joined Toyoda Machine Works, Ltd., Kariya, Japan, where he is a Manager in the New Business Office. His research interests are grinding machines, robotics, intelligent assist devices, and Kansei engineering.



Masashi Yamanaka (M'94) received the B.S., M.S., and Ph.D. degrees from Tohoku University, Sendai, Japan, in 1986, 1988, and 1995, respectively.

From 1988 to 1997, he was an Engineer, working on the development of machine tools, with Toyoda Machine Works, Ltd. In 1997, he joined Tohoku University, where he is currently an Assistant Professor in the Department of Machine Intelligence and Systems Engineering. His current research interests are evaluation of high-performance transmissions, application of traction drives, and intelligent designs for

machine tools.